Bank Data Presentation

Identify Top customers & finding Trends and Patterns

Top Loyal Customers

To find out the Top 10 Customers. I have used the Window function Row_Number() to get my rows arranged in the sequential format like 1, 2, 3, ... Along with this I have used the Over clause to find value in all the rows of my window.

Who are the Top 10 customers -

1. Customers who are having there active account with us.
2. Customers who have High Balance amount.
3. Customers who have a very good Credit Score
4. Customers who have long tenure with our Bank.

Top Loyal Customers

Query | Query History

```sql
-- Finding top 10 loyal customers who have High balance and active with the services :-

SELECT "CustomerId", "Geography", "Surname", "Balance", "CreditScore", "Tenure"
FROM (
SELECT
"CustomerId", "Geography", "Surname", "Balance", "CreditScore", "Tenure", ROW_NUMBER()
OVER(ORDER BY "Balance" DESC) as customer_rank
FROM customer_churn
WHERE "Exited" = 0 AND "IsActiveMember" = 1) AS ranked_table
WHERE customer_rank <= 10;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	CustomerId integer	Geography text	Surname text	Balance numeric	CreditScore integer	Tenure integer
1	15769818	France	Moore	212778.2	850	3
2	15690589	France	Udinesi	212314.03	541	9
3	15795298	Germany	Olisaemeka	206868.78	573	9
4	15668818	Spain	Chidubem	200322.45	592	2
5	15594270	France	Biryukov	198338.77	693	7
6	15666559	Germany	Gould	197715.93	608	8
7	15664506	Spain	Goodwin	197436.82	675	8
8	15614716	France	Okwudilichuk...	196853.62	515	0
9	15605672	France	Yuan	195926.39	694	5
10	15813741	Spain	Nnachetam	193858.2	549	6

Trends and Patterns

Trend 1 : Our 50% of top 10 customers are from “France”, followed by “Spain” and then “Germany”.

Pattern :

- This shows the pattern that also links with the records that Germany have the highest churned rate overall.
- Comparatively less top customers are also from Germany.

Trend 2 : All the top customers have Balance amount ranging from 1 Lakh to 2.2 Lakh.

Pattern :

- It means the highest amount of Balance is 2.2 Lakh.
- Balance pattern is quite constant, no staircase of significant outliers are present even in our top tier lists.

Trends and Patterns

Trend 3 : All the top customers have the CreditScore more than 500.
Pattern :

- This shows the pattern that customers who have very good balance are also managing to have a very good CreditScore.
- 850 being the highest CreditScore, there are minimum a 100 value difference, means CreditScore appears to be a staircase.

Trend 4 : 70% of the top customers have a tenure more than 5 years
Pattern :

- This shows the pattern that top customers are maintaining balance and CreditScore from average of 5 years.
- Unfortunately there is only one customer with 9 years of loyalty, can be consider as an outlier.

Characteristics of the Top 5 Customers

Criteria for the Top 5 customers who are most deserving ones and can get the special services like reduced interest rate, special offers and more premium banking services from us :-

1. Customers who are having there active account with us.
2. Customers who have High Balance amount.
3. Customers who didn’t churned or exited.
4. Customers who have best credit scores.
5. Customers who are using our Credit Card as well.
6. Customers who sticked with us from long time (Tenure), means they are actually very loyal with us so far.
7. Customers who are using more number of Products from our bank services.

Top 5 Customers (Surnames)

01 Hansen

02 Davidson

03 Speth

04 Ward

05 Lee

Top 5 Customers (Details)

Data Output Messages Notifications

Showing rows: 1 to 5

	vip_rank bigint	Surname text	Geography text	CreditScore integer	Balance numeric	Tenure integer	NumOfProducts integer	EstimatedSalary numeric	final_score numeric
1	1	Hansen	Germany	850	141827.33	9	2	149458.73	1331.29
2	2	Davidson	France	850	131996.66	6	2	178747.43	1320.74
3	3	Speth	France	819	122334.26	9	2	181507.44	1312.84
4	4	Ward	France	850	169445.4	5	1	186335.07	1305.78
5	5	Lee	Germany	807	141069.18	1	3	194257.11	1302.33

Identify focus areas and Reasons for customer churn

Customer Churn Analysis (1)

Stat : **Churn peaks at 56% for customers in their 50s.**
Reasons :

- They are likely reaching level of wealth or nearby retirement
- they require Wealth Management or Investment Services that we aren't providing effectively
- leading them to move to "Premium" competitors.

Stat : **Customers with 3 or 4 products churned at a nearly 100% rate.**
Reasons :

- Maybe because more products likely means higher fees or confusing service.
- Probably these customers feel "over-banked" and frustrated, rather than rewarded for their deep relationship with us. This can be the churn rate reason.

Customer Churn Analysis (2)

Stat : **Our German customers are leaving at twice the rate (32%) of those in France or Spain**

Reasons :

- Probably the local service failure means either our competitors in Germany are offering significantly better interest rates
- Maybe our service model doesn't align with German customer's expectations.

Stat : **Many exited customers had 2+ lakh balance and 600+ CreditScore, and tenure of almost 3+ years.**

Reasons :

- It seems that our service policy for these kind of precious customers are not upto the mark, need to hold those valuable customers.
- Even after maintaining high balance and credit-score, they didn't got the significant offers or valuable feelings to stuck for more.

Reasons for choosing the data points

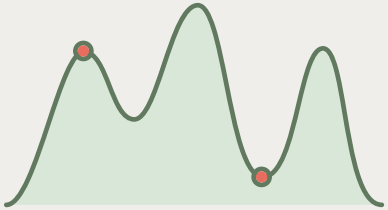
- After looking at 'typical' customers, We found that while a middle-of-the-road customer has nearly ***97,000 (Median)***.
- Our average was being pulled down by ***36%*** of customers having a zero balance, that’s why we didn’t included mean but median.
- This told us we have a huge segment of ***"ghost accounts"*** who are at the highest risk of leaving because they have no interest in our bank.
- We have already done EDA and found insights by univariate and multivariate analysis.
- In one column we didn't just look at age, we looked at ***Wealthy Seniors vs Wealthy Youth.***
- We didn't just look at salaries, we looked at High Earners with only 1 Product vs High Earners with exited status.
- This revealed that our biggest losses aren't random they are predictable. We have found insights previously and got an overall details of what is the problem rate, where are the lacking points, and how to solve it.

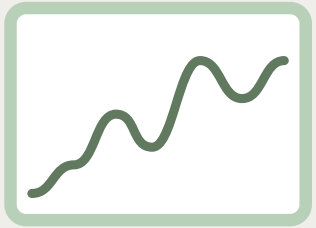
Top 3 Focus Areas

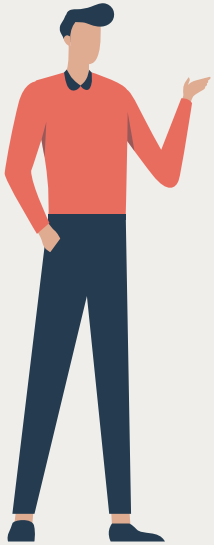
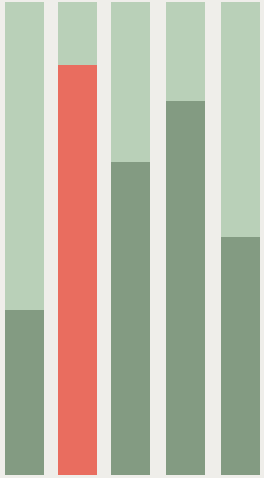
1. Customers aged 45–60 with balances over $100,000 are valuable customers, Offer them VIP or Priority Banking services or reduced interest rates on loans. We cannot afford to lose the segment that holds the most capital.

2. The 36% of customers with a $0 balance. These are "Inactive" members. We need to "activate" them by offering rewards for their first 5 transactions. We can offer them attractive services and call them frequently to make them active customers.

3. Customers holding 2+ products. We need to investigate why people with 3-4 products are leaving. We should simplify the fee structure for multi-product users. We need to change the product’s policy so that our each product justify the value and satisfy our customers on every aspect.







Thank you